

Media Bias ISM

RQ Methods Report

RQ1

Question. How has media reporting bias toward S&P 500 firms changed after the 2020 recession shock?

Variable Construction

For article a of firm i on date t , with MTSC probabilities (p_a^+, p_a^-, p_a^0) , keep high-confidence observations only:

$$\max\{p_a^+, p_a^-, p_a^0\} \geq \tau, \quad \tau = 0.90 \text{ (default)}.$$

Define article stance component and firm-day outcome:

$$s_a = p_a^+ - p_a^-, \quad \text{bias}_{it} = \frac{1}{|\mathcal{A}_{it}|} \sum_{a \in \mathcal{A}_{it}} s_a,$$

where $\mathcal{A}_{it}$ is the set of qualifying articles for firm i on day t .

Controls:

- $\text{vol}_{it} = |\mathcal{A}_{it}|$ (firm-day article volume)
- vix_t (daily market uncertainty)

Shock indicator:

$$\text{Post}_t = \mathbb{I}[t \geq 2020-01-01].$$

Baseline Model (Firm FE)

$$\text{bias}_{it} = \alpha_i + \beta \text{Post}_t + \gamma_1 \text{vol}_{it} + \gamma_2 \text{vix}_t + \varepsilon_{it}.$$

β is the main estimand: average within-firm post-2020 shift in stance.

Estimator and Inference

Within transform by firm means, then OLS:

$$\hat{\theta} = (\tilde{X}^\top \tilde{X})^{-1} \tilde{X}^\top \tilde{y}, \quad \theta = (\beta, \gamma_1, \gamma_2)^\top.$$

Robust HC1 covariance:

$$\widehat{\text{Var}}(\hat{\theta}) = \frac{n}{n-K} (\tilde{X}^\top \tilde{X})^{-1} \left(\sum_{j=1}^n \hat{u}_j^2 \tilde{x}_j \tilde{x}_j^\top \right) (\tilde{X}^\top \tilde{X})^{-1}.$$

Two-sided normal-approx p-values:

$$p_k = 2 \Phi(-|t_k|), \quad t_k = \frac{\hat{\theta}_k}{\widehat{\text{SE}}(\hat{\theta}_k)}.$$

Identification Conditions

1. Conditional within-firm parallel trends after controls.
2. Controls are not post-treatment bad controls in the identification sense.
3. **Stable measurement process:** NewsMTSC score mapping is comparable over time.

Pre-Period Diagnostic and Fallback Trigger

Compute firm pre means:

$$\mu_i^{\text{pre}} = \frac{1}{T_i^{\text{pre}}} \sum_{t \in \text{pre}} \text{bias}_{it}.$$

Track min, max, range, and standard deviation across firms. If range exceeds tolerance (default 0.02), run SCDiD fallback.

SCDiD Fallback (Compact)

- For each firm i , estimate donor weights from pre period:

$$\hat{w}_i = \arg \min_w \|y_{\text{pre}} - X_{\text{pre}} w\|_2^2,$$

then truncate negatives and renormalize.

- Build post synthetic path $\hat{y}_{it}^{\text{syn}} = X_{\text{post},t} \tilde{w}_i$.
- Firm effect:

$$\hat{\tau}_i = \frac{1}{T_i^{\text{post}}} \sum_{t \in \text{post}} (y_{it} - \hat{y}_{it}^{\text{syn}}).$$

- Aggregate fallback effect:

$$\hat{\tau}_{\text{SCDiD}} = \frac{1}{N^*} \sum_{i=1}^{N^*} \hat{\tau}_i, \quad \widehat{\text{SE}} = \frac{s_{\hat{\tau}}}{\sqrt{N^*}}.$$

Fallback Eligibility

A firm is included in fallback only if it has:

- at least 180 observed pre-period days,
- at least 180 observed post-period days,
- sufficient donor coverage for synthetic fitting.

Results

Sample and diagnostics. The estimation panel contains 31,575 firm-day observations across 22 firms (2015-01-01 to 2025-11-12). Pre-period firm means are highly dispersed (range = 0.733, SD = 0.157), so the SCDiD fallback is activated.

Table 1: *

Table RQ1-A: Entity FE Regression (HC1 Robust SE)

Variable	Coef.	SE	t-stat	p-value	CI Low	CI High
Post	0.0258	0.0052	4.9425	0.000001	0.0155	0.0360
Article Volume	-0.0028	0.0003	-9.4603	0.000000	-0.0033	-0.0022
VIX	0.0005	0.0004	1.4886	0.1366	-0.0002	0.0012

Table 2: *

	Metric	Estimate	CI Low	CI High
Table RQ1-B: SCDiD Fallback Summary	Avg. Post Effect	-0.1221	-0.2034	-0.0407
	SE Across Firms	0.0415	—	—
	Firms Used	16	—	—

Interpretation. The baseline FE model indicates a positive post-2020 shift in daily stance. However, the fallback SCDiD estimate is negative, indicating direction sensitivity under strong pre-period imbalance. Report RQ1 as evidence of a structural shift with specification-dependent sign.

RQ2

Question. Did the 2020 recession shock affect stock prices differently pre versus post, at the market level?

Design Rationale

RQ2 is paired with RQ1. RQ1 tests whether media stance shifts after the shock, while RQ2 tests whether firm-level returns also shift. Both are required before claiming a structural media–market linkage in RQ3.

Variable Construction

For firm i and date t , daily return is computed from close prices:

$$\text{return}_{it} = \frac{P_{it} - P_{i,t-1}}{P_{i,t-1}}.$$

Shock indicator:

$$\text{Post}_t = \mathbb{I}[t \geq 2020-01-01].$$

Controls:

- sp500_return_t : absorbs market-wide bull-run and broad index drift.
- vix_t : absorbs volatility-regime shifts common to all firms.

Baseline Model (Firm FE)

$$\text{return}_{it} = \alpha_i + \beta \text{Post}_t + \gamma_1 \text{sp500_return}_t + \gamma_2 \text{vix}_t + \varepsilon_{it}.$$

As in RQ1, estimation uses entity fixed effects and HC1 robust standard errors.

Identification and Fallback Rule

Pre-period firm means are evaluated as in RQ1. If cross-firm pre means are sufficiently equal, FE remains the primary estimate; otherwise SCDiD fallback is triggered.

Results

Sample and diagnostics. The estimation panel contains 62,667 firm-day observations across 26 firms (2015-12-01 to 2025-11-24). Pre-period means are tightly clustered (range = 0.00193, SD = 0.00042), so fallback is **not** triggered.

Table 3: *
Table RQ2-A: Entity FE Regression (HC1 Robust SE)

Variable	Coef.	SE	t-stat	p-value	CI Low	CI High
Post	-0.000344	0.000172	-2.0038	0.0451	-0.000681	-0.000008
S&P 500 Return	1.127576	0.010982	102.6767	0.0000	1.106052	1.149100
VIX	-0.000001	0.000020	-0.0433	0.9654	-0.000039	0.000038

Interpretation. Controlling for index return and VIX, the post-period shift in firm daily returns is negative and marginally significant (about -0.034% per day). The market-return control is strongly positive and highly significant, while VIX is not statistically significant in this specification.